

MSE 311 Thermodynamics of materials-Quiz 6

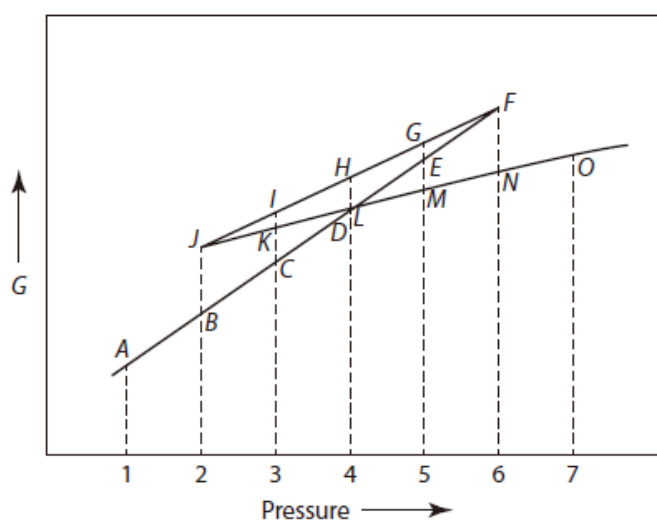
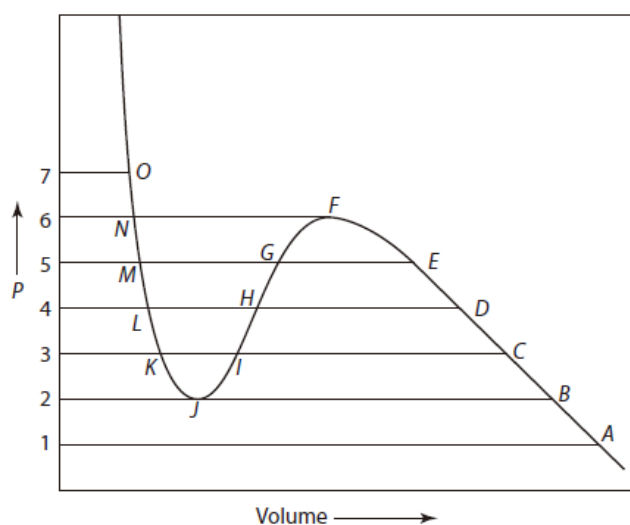
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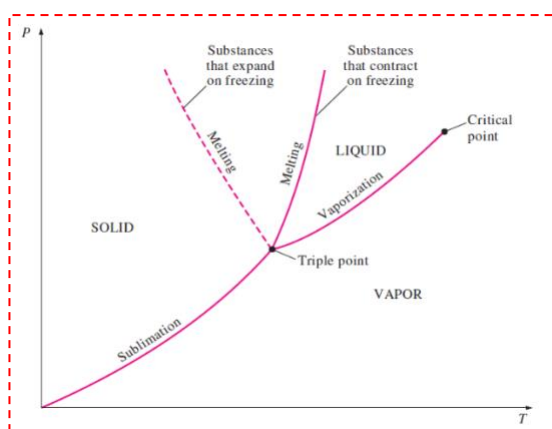
1. Write down the *van der Waals equation* for n mole gas.

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

2. Draw the change of Gibbs free energy versus pressure from state A to state O according to the left figure.



2. Draw the melting curve, vaporization curve and sublimation curve of water in the following figure, and indicate the phases (solid, liquid and vapor) of the water.



3. Write the relationship between the pressure and temperature according to *Clausius-Clapeyron equation* considering ΔH is independent or dependent ($\Delta H = -AR + BRT$) of temperature.

Clausius-Clapeyron equation: $d \ln P = \frac{\Delta H}{RT^2} dT$

1) If ΔH is independent of temperature:

$$\ln P = -\frac{\Delta H}{RT} + C$$

2) If ΔH is dependent of temperature: $\Delta H = -AR + BRT$

$$d \ln P = \frac{-AR + BRT}{RT^2} dT = \left(-\frac{A}{T^2} + \frac{B}{T} \right) dT$$

$$\ln P = \frac{A}{T} + B \ln T + C$$